# Venkatesh D

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## SUMMARY

MSc. Data Science, Coimbatore Institute of Technology. working with python, sql, scikit-learn, pytorch, feature engineering, data modeling.

## SKILLS

python · sql · scikit-learn · pytorch · feature engineering · data modeling · dax · eda · excel · flask · git · gnn · mongodb · mysql · neo4j · opencv · postgresql · power bi · power query · tensorflow

## EXPERIENCE

**Unirols Airtex – Web Development & AI Integration** Dec 2025 - Feb 2026

- Architected a full-stack corporate website using React and Flask, structuring product data for improved navigation and customer engagement. Integrated an AI-based chatbot to automate product inquiries and enhance customer support efficiency.

## PROJECTS

**TRI-BIOMETRIX** *Intelligent Multimodal Biometric Security System LINK* Sep 2025 - Oct 2025

- Tools & Technologies: Python, TensorFlow, OpenCV, Flask
- Implemented a 3-factor authentication system integrating face, voice, and gesture recognition. Applied deep learning models to enhance identity verification accuracy and system reliability.
- Reduced unauthorized access attempts by ~60% through 3-layer biometric validation pipeline through layered biometric validation

**UPI ₹15+ Lakh Crore Digital Payments Velocity Dashboard LINK** Jan 2026 - Feb 2026

- Tools & Technologies: Power BI, DAX, Power Query, SQL, Excel
- Analyzed Rs.15+ lakh crore RBI UPI transaction datasets using SQL and DAX to evaluate payment trends and state-wise growth patterns to deliver KPI-driven insights into transaction velocity and digital market expansion.
- Built interactive a 5-page Power BI dashboard integrating transaction, channel, and customer datasets to uncover behavioural patterns and operational performance metrics.

**NETRATAX – Tax Fraud Detection using GNN LINK**

- Tools & Technologies: Python, SQL, PyTorch, Flask, PostgreSQL, Neo4j Nov 2025 - Dec 2025
- Engineered a data-driven fraud detection system analyzing 500K+ GST transaction records across states to identify shell entities and circular trading patterns. Structured relational datasets and automated graph construction for scalable risk analysis.
- Built fraud risk scoring model achieving 87% accuracy with <5% false positives.
- Delivered 4 interactive dashboards for fraud network visualization and risk-based case prioritization.

## EDUCATION

**MSc. Data Science** *Coimbatore Institute of Technology CGPA: 7.52/10 – up to 5th Sem* 2023 – 2028

**Higher Secondary** *Bala Bharathi Matric Higher Secondary School 90 %* 2022 – 2023

## ACHIEVEMENTS

- SMART INDIA HACKATHON 2025 - Internal Hackathon winner; recognized as top 50 teams.

- TECH SCULPT PROJECT EXPO 2025, IEE STUDENT BRANCH - Winner, awarded for excellence in technological innovation.
- CIT SPARK GRANT 2025 – Finalist among 25 teams; pitched innovative project demonstrating impact and technical excellence.
- ASTRANOVA HACKATHON 2026 (Dept of IT, CIT) – 1 Runner-up, recognized for high-impact innovation.